Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL, TypeScript
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, database design
- **Infrastructure & DevOps:** Docker, AWS, Linux
- **Testing & Validation:** Test case development, testing, validation

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) Jan 2026 – Present

Personal Project · Python, LangChain, Gemini API, Streamlit, RAG Architecture

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot in Python, enabling the discovery, extraction, and parsing of real-time Software Engineering postings.
- Built a zero-hallucination **RAG (Retrieval-Augmented Generation)** engine using LangChain, Gemini APIs, and a sandboxed FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.
- Designed and implemented a Streamlit Human-in-the-Loop dispatch UI with a Kanban board, enabling users to review, tailor, and manage applications across a full pipeline with persistent state via a SQLite repository layer.
- Applied **Test-Driven Development (TDD)** with a 113-test suite using Pytest, mocking, and async fixtures to validate all components including the scraper, AI tailor, PDF generator, and repository layer.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 – Jan 2026

Personal Project · Python, Docker, AWS, FastAPI, SQLModel

- Migrated full-stack platform from Azure to AWS (EC2), implementing **Docker** container orchestration to optimize deployments and reduce infrastructure costs.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed a normalized schema using SQLModel with JSON-type columns to store recursive tree structures, optimizing read performance for complex queries and **database design**.

TitanStore (Distributed Raft Database) Jan 2026 – Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, WAL

- Developed a distributed key-value database in Go from scratch, implementing the **Raft consensus algorithm** for leader election and fault-tolerant log replication across a cluster of nodes.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss.
- Designed and implemented a high-throughput raw TCP server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O.

EDUCATION

Bachelor of Science, Computing Science May 2025 – Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025